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$^{40}Ar/^{39}Ar$ and paleomagnetic results from Liberia and the Precambrian APW data base for the West African Shield

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Abstract—Lower amphibolite to granulite facies metamorphic rocks in Nimba County, Liberia have yielded 2.2–2.9 Ga Rb–Sr whole rock ages, indicating that they are part of the Archean Liberian age province. We report a 2040 Ma $^{40}Ar/^{39}Ar$ plateau date on hornblende from an amphibolite in this region, and suggest that these rocks were also severely reworked during the Eburnean (~2.0 Ga) metamorphic episode. $^{40}Ar/^{39}Ar$ analyses of biotite and feldspars from neighboring schists also indicate the presence of two mild thermal events, at 1.5 Ga and 0.6 Ga. Paleomagnetic analyses of samples from these same metamorphic rocks reveal three components of magnetization. The predominant and most stable component (273°E, 21°N) is considered to have been acquired as a result of post Eburnean uplift and cooling at ~2.0 Ga, whereas the two less stable components with poles at 235°E, 43°N and 16°E, 36°N, probably correlate with the 1.5 Ga and 0.6 Ga thermal pulses, respectively. Rock units from southern Liberia also yield two secondary magnetizations, one at 247°E, 37°N and the other at 104°E, 5°N, and a 1.5 Ga $^{40}Ar/^{39}Ar$ date on plagioclase. Comparison of the paleomagnetic poles corresponding to the ~2.0 Ga Eburnean component with published paleomagnetic data for West Africa is not consistent with prior interpretations of the polar wander path for West Africa. Our paleomagnetic data, when compared to poles of comparable age from the Kalahari Shield, still suggest that some form of displacement has occurred between the Kalahari and West African Shields since 2.0 Ga.

INTRODUCTION

ONSTOTT *et al.* (1984) demonstrated the agreement between contemporaneous 1.95 Ga paleomagnetic poles from the West African Shield in southern Liberia and the Guiana Shield in Venezuela when the Atlantic was closed. This study extends the Liberian paleomagnetic sample base to amphibolites and granulites in Nimba County, northeastern Liberia, to determine whether the magnetizations in these Archean rock units were also reset by 2.0 Ga Eburnean orogeny and, if so, whether they agree with the Eburnean magnetizations reported for southern Liberia.

The new $^{40}Ar/^{39}Ar$ and paleomagnetic results are then compared to previously published paleomagnetic data from other parts of the West African Shield (Piper and Lomax 1973, Sabate and Lomax 1975, Morris and Carmichael 1978, Morel 1981, Onstott *et al.* 1984), in order to refine the West African Precambrian paleomagnetic APW path.

GEOLOGIC SETTING

The basement rock units in this region consist primarily of quartzo–feldspathic gneisses with minor pyroxene–plagioclase gneisses and orthoamphibolites (Fig. 1). These are unconformably overlain in the northeast part of the field area by metasedimentary and metavolcanic rocks of the Nimba Supergroup (Berge 1968). In general, the metamorphic grade increases from greenschist/epidote amphibolite facies in the east, within the Nimba Supergroup, to upper amphibolite/granulite facies in the west and southwest (Berge 1968, Force *et al.* 1971–1972). Two sets of isoclinal folds and one set of open folds appear to predate the Eburnean metamorphism. A fourth set of open upright folds strike northwest–southeast parallel to the trend of the coastal Pan-African belt (Force 1983).

Six Rb–Sr whole rock model dates ranging from 2590 to 3350 Ma, were reported by Hurley *et al.* (1971) for the basement gneisses. Four of these six samples fall on a 2349 ± 80 Ma isochron, with an initial $^{87}Sr/^{86}Sr$ ratio of 0.710 (Hurley *et al.* 1971, least squares fitted by the authors). Two samples of Nimba Supergroup schists yield Rb–Sr model ages of 2040 and 2370 Ma and fall on a 2154 Ma isochron when plotted with analyses of similar schists that occur in Sierra Leone (*op. cit.*). Finally, Hedge *et al.* (1975) report a 2220 ± 60 Ma K–Ar hornblende age from Yekepa mafic basement gneiss. The Rb–Sr ages on the schists and the K–Ar hornblende age suggest that both the strontium and argon systems of these rocks have been disturbed by the 2.0 Ga Eburnean metamorphism.

FIELD AND LABORATORY PROCEDURES

Sixty-one hand samples oriented by magnetic and sun compass were collected from nine localities within

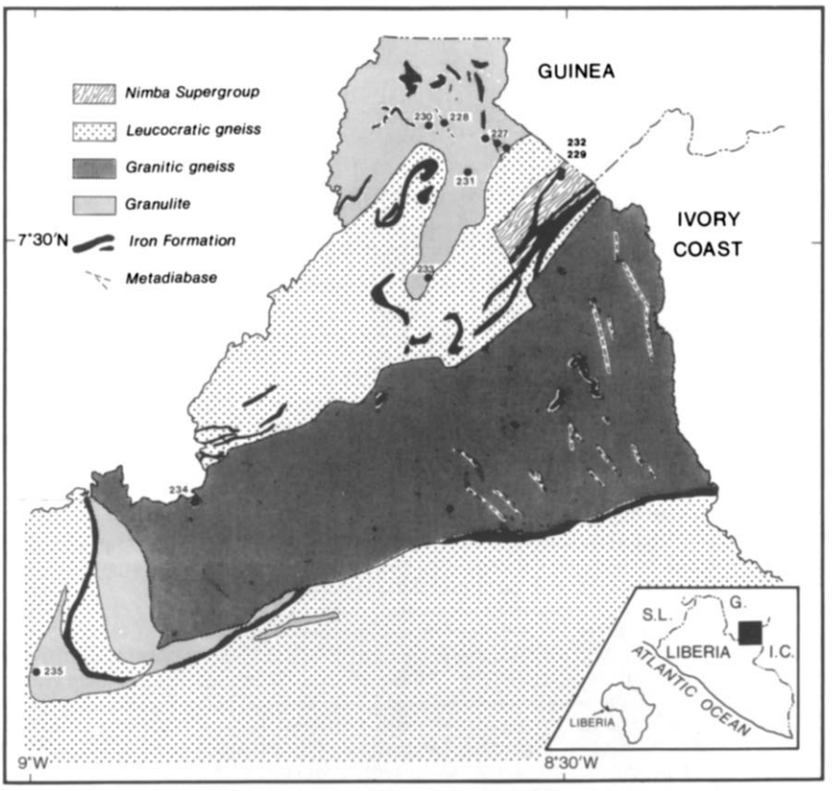


Fig. 1. Basement geology of the northeastern portion of Liberia adapted from Force and Berge (1977). Paleomagnetic sample localities (solid circles) and site numbers are shown. Inset portrays regional location and political boundaries.

Nimba county (Fig. 1). One inch (2.5 cm) cores were drilled from these samples, oriented in the lab, and cut to provide at least two specimens. In the text the number '###.##.##' corresponds to 'site number·sample number·specimen number'.

The remnant magnetizations were measured on a Schonstedt SM-1 spinner magnetometer interfaced to an Apple IIe computer. Alternating field (A.F.) demagnetization was performed with a two axis tumbler in ambient DC fields of less than 30 nT and with peak alternating fields up to 130 mT. The orientation of the specimen in the tumbler was reversed with each consecutive step to correct for any rotation induced remanence (Hillhouse 1977). Thermal demagnetization was performed on a Schonstedt TSD-1 with a measured ambient DC field of 20 nT and a measured accuracy of ±5°C in temperature. Forty-five minute heating intervals were used to ensure samples were in thermal equilibrium with the furnace temperature. This equipment is housed in a double layer, steel room with an ambient field of 1500 nT.

All samples were subjected to detailed A.F. demagnetization in 2.5 mT steps up to 20 mT, and 5 mT steps up to 50 mT, and 10 or 20 mT steps up to 120 mT or until the directions began to scatter. Fourteen samples were stepwise thermally demagnetized in 50°C steps from 200 to 500°C, 10°C steps between 500 and 600°C, and 30°C steps from 600 to 690°C or until the directions scattered.

Mean site directions or convergence directions and corresponding statistics (Fisher 1953, Halls 1976, Onstott 1980), were calculated at the sample level, using only those specimens that satisfied the following minimum acceptance criteria; (1) the least squares lines or remagnetization planes were composed of at least four consecutive demagnetization steps; and (2) the mean angular deviation (MAD) (Kirschvink 1980) of these lines and planes was less than 20°.

Polished thin sections were made from one to three representative samples from each site and examined under transmitted light, and under reflected light using magnetic colloid to aid identification of the opaque mineral assemblages. For one whole rock sample from each site, the saturation magnetization was measured as a function of temperature upon heating in air, utilizing an automated, vertical Curie balance, similar to that designed by Cox and Doell (1967), interfaced to an Apple IIe computer.

Plagioclase and hornblende were separated from an amphibolite sample, and biotite and K-feldspar were separated from an adjacent biotite schist, at site 227, utilizing a Franz magnetic separator and heavy liquids. Hand picking was used in the final stage of sample preparation to ensure 99.9% purity. These samples and three aliquots of irradiation standard, Zartman Hb3gr hornblende (1072 Ma; Turner *et al.* 1971), were irradiated in quartz vials at McMaster University research reactor in Hamilton, Ontario. The standards were evenly distributed in the cannister (40 mm × 19 mm dia.) to average any flux gradients, which generally tend to be less than 1% (Masliwec 1981). The samples were stepwise degassed in a quartz tube by a Lindberg high

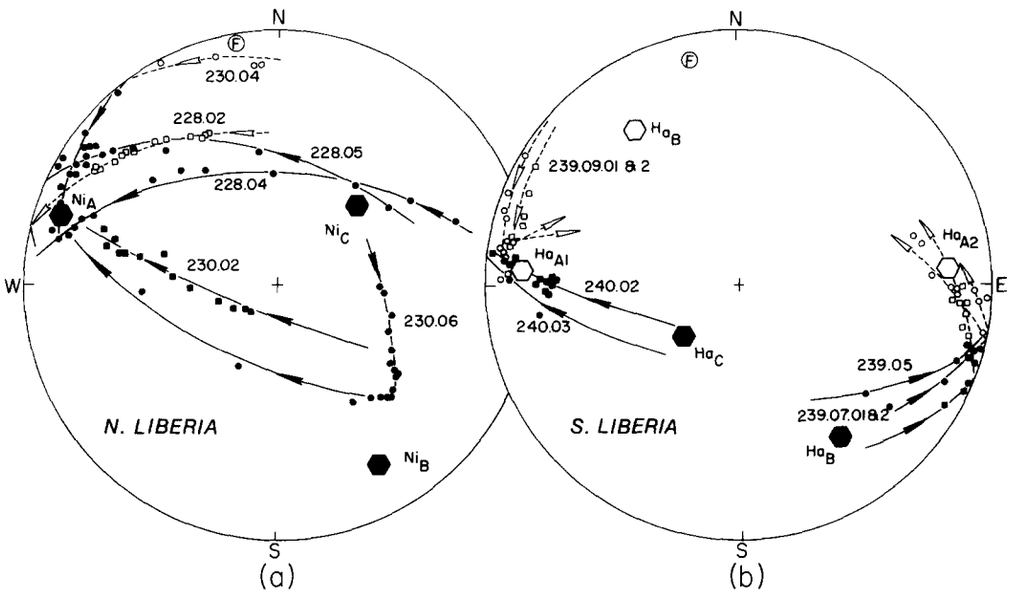


Fig. 2. Stereonet plots for A.F. (circles) and thermal (squares) data and corresponding remagnetization planes for samples from sites 228 and 230 in northern Liberia (a) and sites 239 and 240 in southern Liberia (b). Open circles and dashed lines are upper hemisphere and solid circles and solid lines are lower hemisphere. Arrows point in the direction of higher demagnetization steps. Hexagons represent the least squares estimated convergence direction. 'F' represents the present day local geomagnetic field. Symbols are discussed at appropriate places in the text.

temperature furnace with a system blank of $0.25 \pm 0.04 \times 10^{-10}$ ccSTP ^{36}Ar at 600°C to $0.7 \pm 0.10 \times 10^{-10}$ ccSTP ^{36}Ar at 1200°C. The isotopic composition of the argon was measured with a AEI MS10 at the University of Toronto, and the measurements were routinely corrected for mass discrimination, sensitivity, decay of ^{37}Ar and ^{39}Ar, interference reactions, system blanks, and atmospheric contamination. The errors were evaluated by propagation of peak height errors derived from linear least squares regression on seven measurements of each peak (^{36}Ar through ^{40}Ar) using a program written by Hall (1982). The ages were calculated using the constants of Steiger and Jager (1977). For any sample for which the ages of four or more consecutive temperature steps agree within two-sigma error limits, a plateau date was determined by calculating an integrated date for that portion of the age spectrum on the plateau.

PALEOMAGNETIC AND PETROGRAPHIC RESULTS

Northeastern Liberia

Table 1 summarizes the paleomagnetic results and petrographic observations. Representative demagnetization data are illustrated on orthogonal projections, stereoplots, and intensity plots in Figs. 2–6. For the majority of the samples the intensity of remanent magnetization varies from 0.01 to 0.7 Am^{-1} and the median destructive field (MDF) ranges from 2 to 30 mT. Upon A.F. demagnetization of the 61 samples, five yielded single component magnetizations, whereas 47 yielded multicomponent magnetizations and nine behaved unstably and failed to yield any component that satisfied our minimum acceptance criteria. Eight of the 47 multicomponent samples possessed three components and 39 yielded two component magnetizations. Only 65 of these 102 component directions from the multicomponent samples and all five single component samples satisfied our minimum cirteria of acceptability (see Table 1). Based upon the A.F. demagnetization data, the following three characteristic magnetizations, Ni_A, Ni_B and Ni_C, were identified:

Component Ni_A. A westerly, positive inclination direction is present in 28 samples (5 single component, 23 multicomponent) at five sites (227, 228, 230, 234 and 235). In multicomponent samples the remanent directions converge upon the Ni_A direction as A.F. and thermal demagnetization preferentially removes less stable components Ni_B and Ni_C and present day field viscous overprints (Fig. 2a). The Ni_A component survives peak fields of 120 mT in the case of the amphibolite at site 228 (sample 228.05.02 in Figs. 3b,d) and neighboring pyroxene–granulite (samples 230.01.02 and 230.02.01 in Fig. 4), and peak fields of 50 mT in the case of the granitic gneiss at site 234 (sample 234.01.12 in Fig. 5a,c) and the metanorite at site 235 (sample 235.07.02 in Fig. 5b,d). With thermal demagnetization, the Ni_A component was consistently removed between 550 and 580°C (see Figs. 3c,d, 4c,d, and 5c,d), suggesting that single-domain or pseudo-single-domain magnetite is the dominant carrier of component Ni_A.

The metanorite contained large grains of orthopyroxene rimmed by clinopyroxene and hornblende with a texture suggesting that the orthopyroxene had

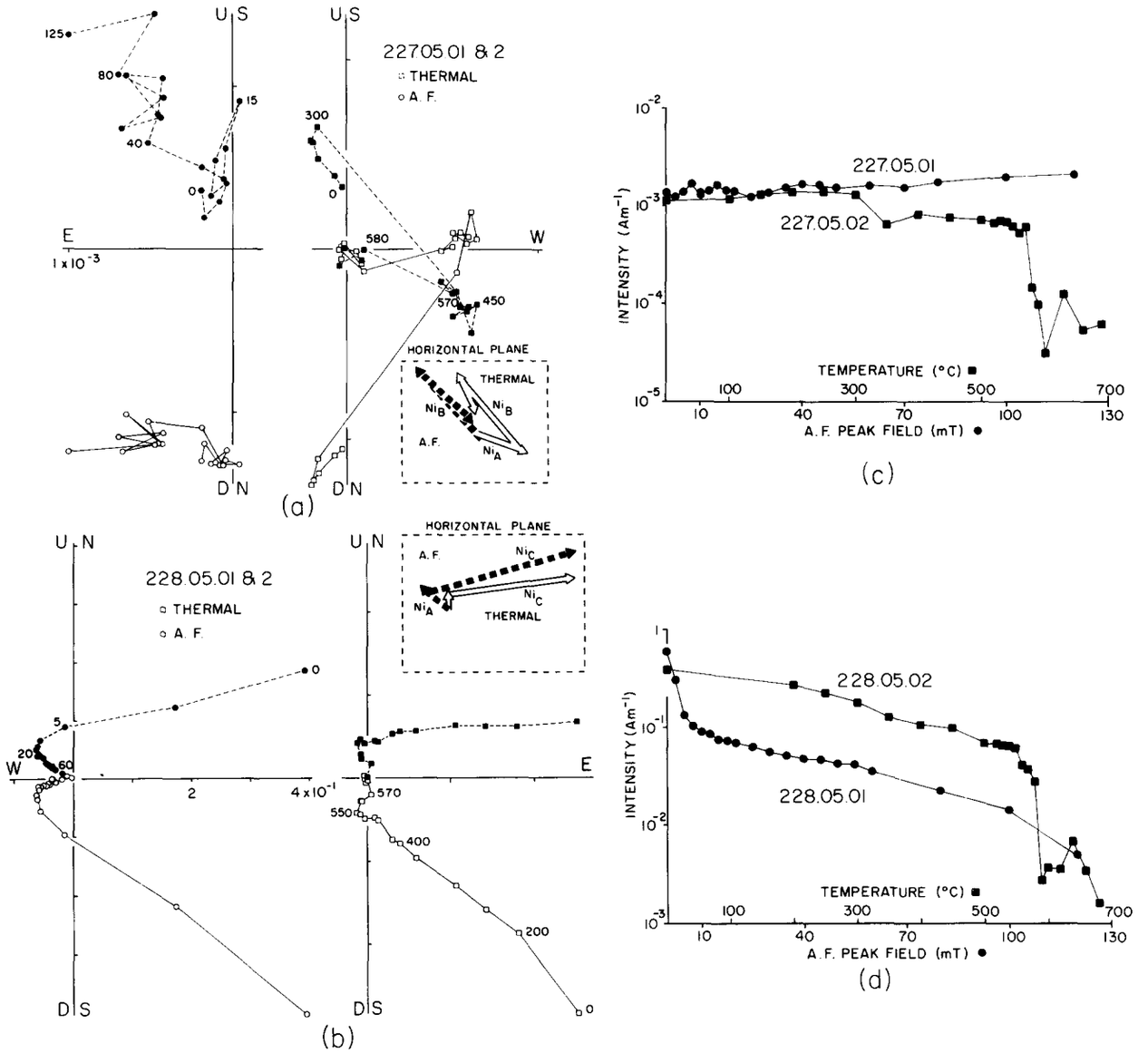


Fig. 3. Orthogonal plot of A.F. (circles) and thermal (squares) demagnetization of (a) amphibolite sample 227.05 and (b) amphibolite sample 228.05. The horizontal projection is plotted with solid symbols and dashed lines, whereas the vertical projection is plotted with open symbols and solid lines. The inset box contains a cartoon portrayal of the authors' interpretation of the components removed by thermal (solid open arrows) and A.F. (dashed arrows) demagnetization. Intensity is in S.I. units (Am^{-1}).

undergone late stage deuteric or metamorphic alteration. The orthopyroxene was riddled with fine-grained, needle-shaped inclusions, assumed to be magnetite, not present in the rimming clinopyroxene or hornblende, suggesting that they formed either before or during the post-metamorphic retrogression. As revealed by magnetic colloid, extremely narrow (~0.5 μm) lamellae of magnetite within otherwise homogenous, ilmenite were also present (magneto-ilmenite). Both the inclusions and the magnetite lamellae are candidates for the carriers of the SD-magnetite remanence observed in the metanorite. Accessory ilmeno-magnetites (~50 μm) with sandwich lamellae were observed in sections of samples from sites 228 and 234. Relict ilmeno-magnetites almost entirely converted to hematite and rutile were prevalent in sections of granulite from site 230. Fine-grained opaque inclusions in silicates (i.e. pyroxene, plagioclase or hornblende), which could possibly represent single-domain magnetite, were not observed in any samples from sites 227, 228, 230 and 234. Based upon the remanent demagnetization properties, however, we presume that sub-microscopic, single-domain, magnetite inclusions do exist, at least for sites 227, 228 and 230.

Component Ni_B. This predominantly southeasterly, positive inclination component is reported for six samples from site 227, three from site 230, and two from site

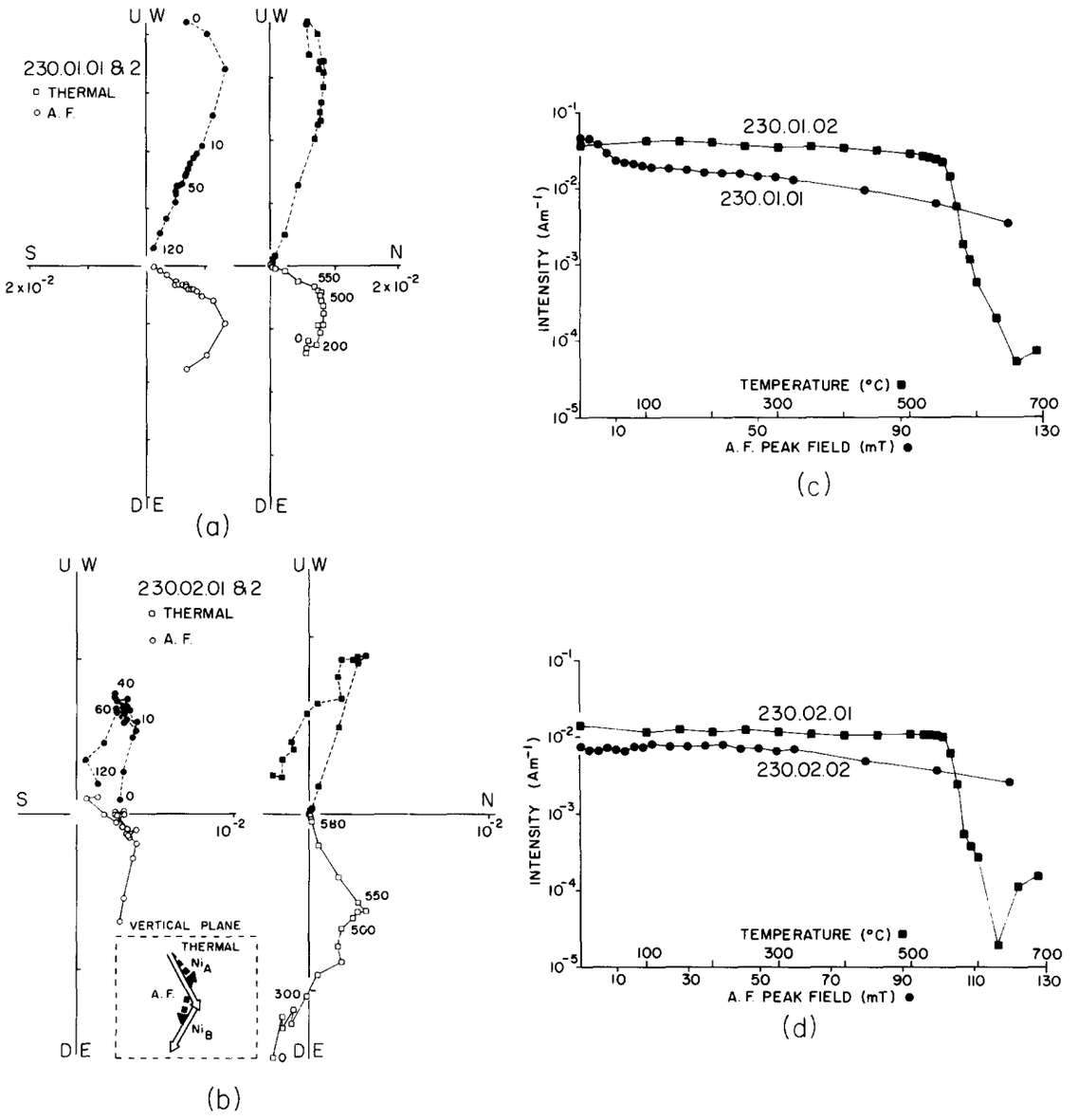


Fig. 4. Orthogonal plot of A.F. (circles) and thermal (squares) demagnetization of granulite samples (a) 230.01 and (b) 230.02. Symbols are the same as in Fig. 3.

228. It is present at other sites, but at these it does not meet our minimum criteria of acceptance. In site 227 samples, Ni_B is very stable to A.F. demagnetization (see Fig. 3a), but is completely removed by thermal demagnetization between 300 and 350°C. This sharp unblocking temperature plus high stability to A.F. demagnetization suggest that Ni_B probably is carried by pyrrhotite, which is indeed observed in sections of site 227 samples. In the pyroxene bearing granulite at site 230, Ni_B is less stable to A.F. demagnetization than Ni_A, being erased between 5 and 10 mT. With thermal demagnetization, Ni_B is partially removed by 400°C, followed by removal of a mixture of Ni_B and Ni_A up to 520°C, above which only Ni_A remains (Figs. 2 and 4b). In this case, the Ni_B component probably resides in the very minor amount of multidomain magnetite, which together with ilmenite, appears to be associated with alteration of biotite in the granulite at site 230.

Component Ni_C. This predominantly northeasterly, steep, positive inclination direction was listed in only 16 samples from sites 227, 228, 230, 232 and 234. Although present in other samples from other localities, in these the data do not meet our minimum criteria for acceptance. In samples from the amphibolite at site 228 (Fig. 3b), and the granitic gneiss at site 234 (Fig. 5a), the Ni_C component is generally removed by A.F. demagnetization between 5 and 7.5 mT. Ni_C is preferentially removed by A.F. demagnetization with respect to Ni_B and Ni_A from granulite sample 230.06 (Fig. 2a), which contains all three components. It is only partially removed, however, with thermal demagnetization at

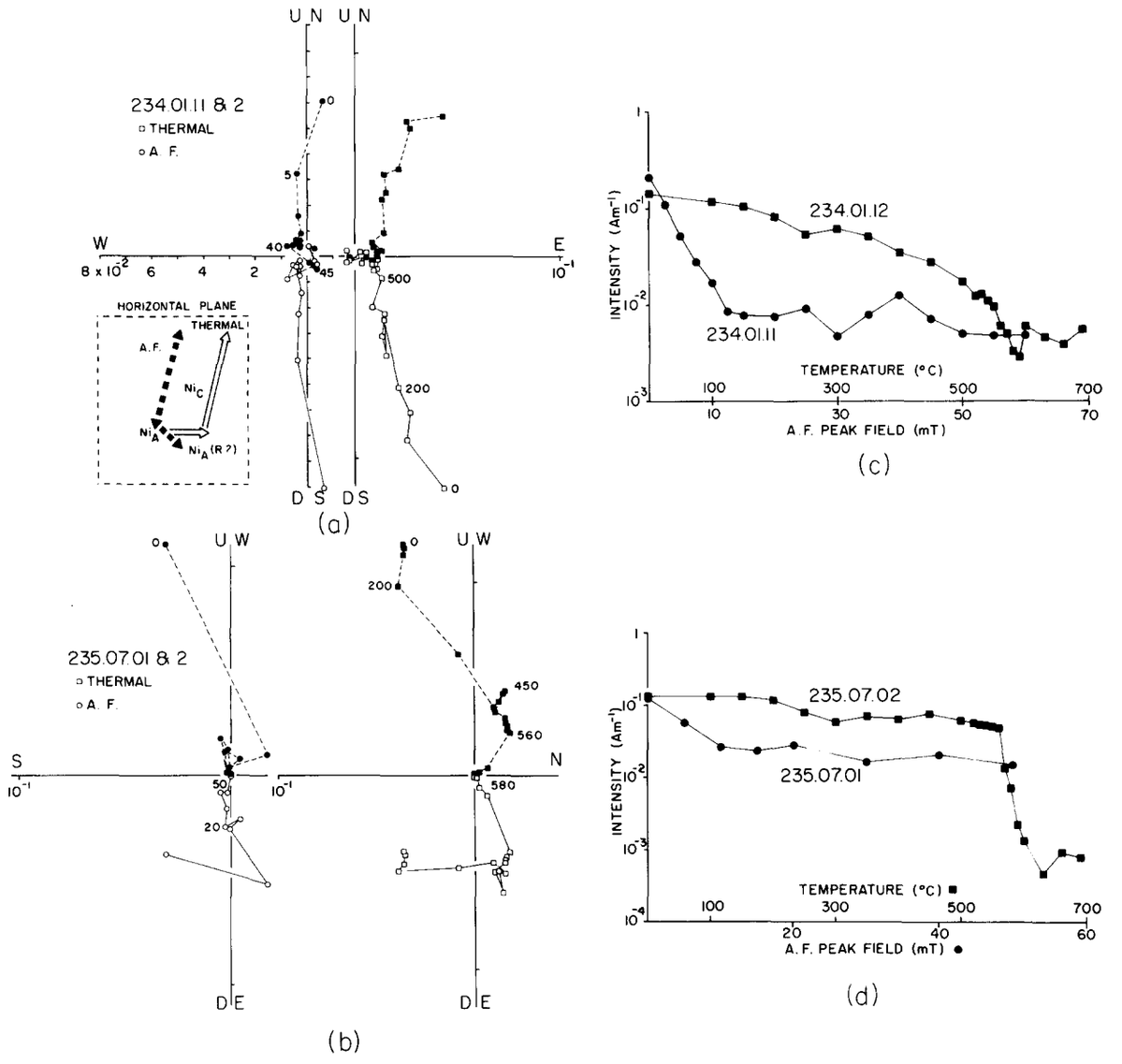


Fig. 5. Orthogonal plots of A.F. (circles) and thermal (squares) demagnetization of (a) granitic gneiss sample 234.01 and (b) metanorite sample 235.07. Symbols are the same as in Fig. 3.

500–540°C, above which both Ni_C and Ni_A are removed simultaneously. In these cases, we suspect that multidomain magnetite is the carrier of Ni_C, because of its low coercivity, relatively high unblocking temperature, and its petrographically observed presence. In samples from site 232, however, Ni_C is stable up to 50 mT, but completely removed by thermal demagnetization between 250 and 300°C (Fig. 6a,c). These properties, combined with the observed abundance of pyrrhotite in thin section in this low grade schist, suggest that pyrrhotite is the carrier of component Ni_C.

The remaining samples from four sites (229, 231, 232, 233) yield highly dispersed directions (Fisher's $\kappa \leq 4$) and low MDFs ranging from 3 to 7 mT and are characteristically unstable at A.F. demagnetization above 15 mT and thermal demagnetization above 300°C. Three of the sites (231, 232 and 233) are weakly magnetized (<0.1 Am^{-1}) and contain abundant, large grains ($>100\,\mu$m) of pyrrhotite and ilmenite to the exclusion of magnetite. One sample from site 231 reveals a downward deflection in the saturation magnetization at ~300°C (Fig. 7), close to the Nèel temperature of pyrrhotite. The most intensely magnetized site of the four, 229, contains abundant multidomain magnetite, which is severely altered to chlorite and rutile.

Southern Liberia

Paleomagnetic $^{40}Ar/^{39}Ar$ data for amphibolites in southern Liberia near the town of Harper were reported by Onstott *et al.* (1984). In this section, we present additional $^{40}Ar/^{39}Ar$ data and the results from a more detailed analysis of the paleomagnetic data.

Of the 25 samples collected from three sites, 19 yielded two components of magnetization, four gave a single

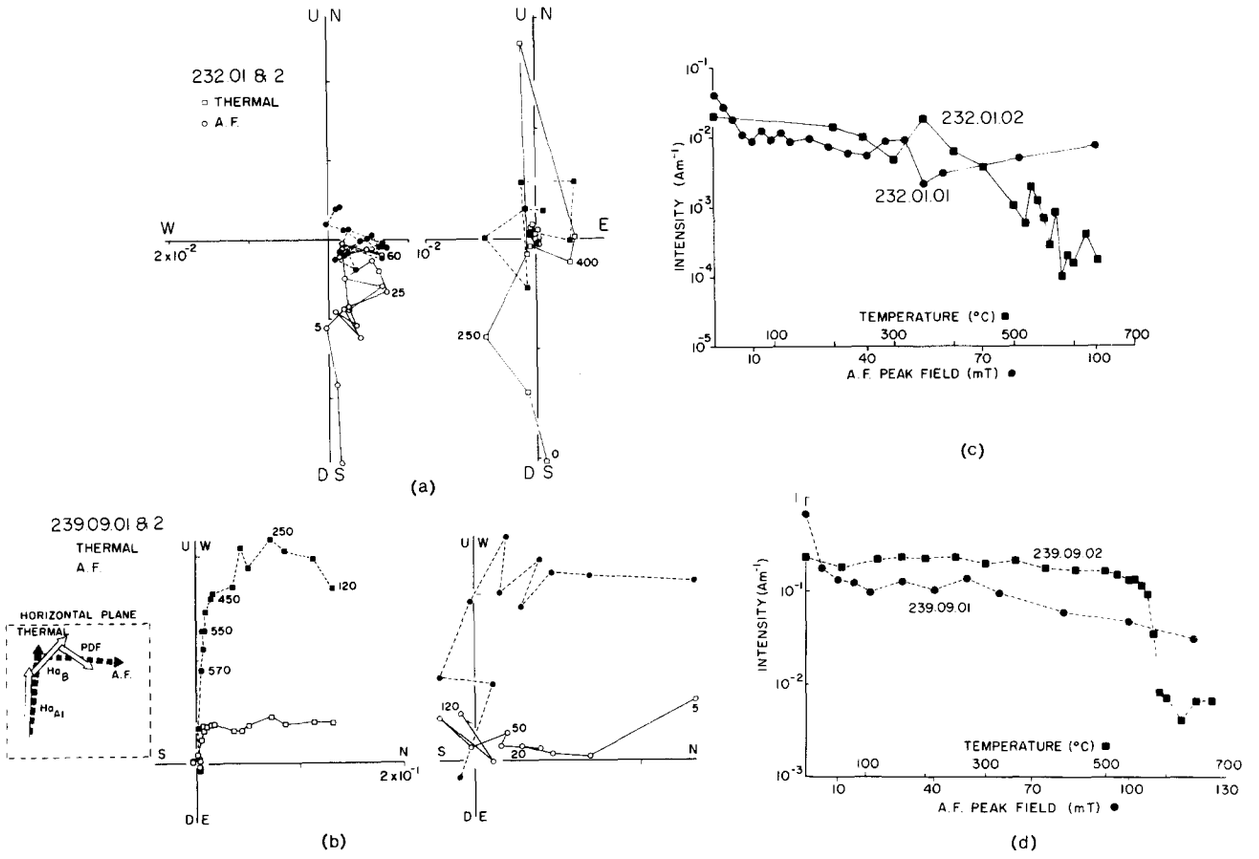


Fig. 6. Orthogonal plots of A.F. (circles) and thermal (squares) demagnetization of (a) amphibole schist sample 232.01 and (b) amphibolite sample 239.09. Symbols are the same as in Fig. 3.

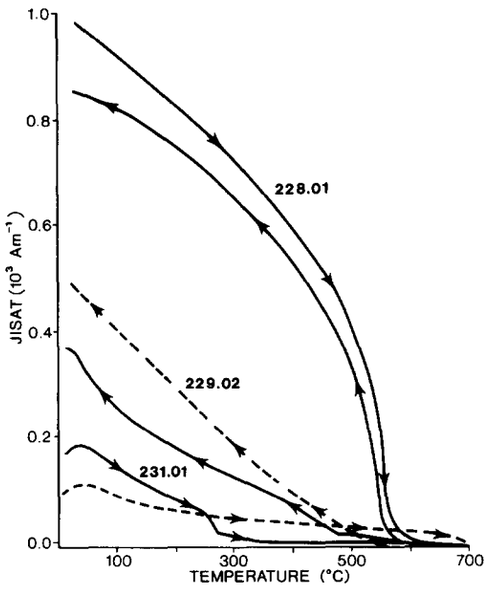


Fig. 7. Saturation magnetization (Am^{-1}) versus temperature curves for samples 228.01 (solid line), 229.02 (long dashed line), and 231.01 (short dashed line) heated in air. Arrows indicate the direction of heating and cooling.

remanent vector, and two did not yield stable results upon A.F. demagnetization. The predominant and most stable component is a westerly, shallow inclination direction found in 12 samples and present in both polarities, which Onstott *et al.* (1984) referred to as Ha_{A1} and Ha_{A2}. In the multicomponent samples, the remanent directions converge upon either Ha_{A1} or Ha_{A2} with A.F. and thermal demagnetization (Fig. 2b) preferentially removing two other components, Ha_B and Ha_C.

Component Ha_B consists of northwesterly, negative inclination (Fig. 6b,d) and southeasterly, positive inclination directions, which are present in 11 samples (see Table 1). Ha_B is less stable to A.F. demagnetization than the Ha_{A1} or Ha_{A2} components, usually requiring 15–30 mT for removal and is typically erased with thermal demagnetization between 350 and 450°C (Fig. 6b).

Component Ha_C consists of southwesterly, steep, positive inclination directions and is present in 14 samples, eight meeting our minimum criteria of reliability. Ha_C tends to be less stable to A.F. demagnetization than either the Ha_A or Ha_B magnetizations, usually requiring 8–20 mT for removal, and is erased by 250–300°C with thermal demagnetization.

Those samples in which Ha_B or Ha_C is the predominant component, contain abundant large grains (>100 μm) of homogeneous magnetite, suggesting that both are associated with multidomain magnetite.

Table 1. Paleomagnetic data for rock units from Liberia, West Africa

Rock type	Site No.	Site lat. (°N)	Site long. (°W)	N	INT (Am^{-1})	MDF (mT)	OA
Northeastern Liberia							
Granitic gneiss and amphibolite	227	7.55	8.55	9	0.111	29	py, h
Amphibolite dike	228	7.55	8.60	5	0.536	12	m, i i–m, py
Nimba Supergroup amphibolite	229	7.51	8.51	6	0.392	3	m
Pyx. granulite	230	7.55	8.60	6	0.136	24	i–m + h–r
Amphibolite	231	7.51	8.55	4	0.0243	3.3	po, i, py
Nimba Supergroup Amphibole schist	232	7.50	8.50	5	0.0132	4.3	i, po
Amphibolite	233	7.50	8.50	5	0.0842	7.2	po, i, py
Granitic gneiss	234	7.25	8.85	6	0.223	2.3	i–m
Metanorite	235	7.10	9.00	15	0.709	2.3	m–i, po, oi
Southern Liberia							
Amphibolite	239 240 241	4.40	7.70	25			

Sites averaged
227, 228, 230, 234 and 235
227, 228 and 230
227, 228, 230, 232 and 234
227, 228, 230, 234, 235 and Ha_{Al}
227, 228, 230 and Ha_B
227, 228, 230, 232, 234 and Ha_C

N = number of samples collected and included in intensity and MDF averages.
INT = NRM intensity in Am^{-1}.
MDF = medium destructive field in mT.
OA = opaque mineral assemblage observed under reflected light microscopy.
i = homogeneous ilmenite; i–m = magnetite with ilmenite lamellae; m–i = ilmenite with magnetite lamellae; m = homogeneous magnetite (multidomain); h = hematite; h–r = hematite–rutile intergrowth; py = pyrite; po = pyrrhotite; oi = opaque inclusions in silicates. Phases are listed in order of abundance.
n = Number of samples included in site average directional statistics based on minimum criterion discussed in text.
Comp. = Classification of magnetic components.
T = Demagnetization treatment thermal (T) and alternating field (A.F.).
Dec., Inc. = Site average direction.
κ = Fisher's (1953) concentration parameter.
κ_1/κ_2 = Bingham's concentration parameters.
α_{95} = Fisher's (1953) 95% confidence cone.
α_1/α_2 = Bingham's 95% confidence ellipses.
* = Data random at 95% level according to the Bingham uniformity test.
† = From Onstott *et al.* (1984).

n	Comp.	T	Dec.	Inc.	κ κ_1/κ_2	α_{95} α_1/α_2	Pole long. (°E)	Pole lat. (°N)
5	Ni_A	A.F.	282.7	21.4	9	27.2	271.8	14.3
6	Ni_B	A.F.	152.2	19.6	11	21.5	228.3	56.0
1	Ni_C	A.F.	76.8	83.9	—	—	3.3	10.2
3	Ni_A	A.F.	286.1	8.3	21	27.6	263.6	16.4
2	Ni_B	A.F./T	308.2	−26.7	41	39.9	238.3	34.4
3	Ni_C	A.F.	43.8	53.2	9	44.7	36.4	43.2
4	?	A.F.	42.3	37.8	3*	72.1		
4	Ni_A	A.F.	287.9	8.5	98	9.3	263.4	18.3
3	Ni_B	A.F./T	133.1	29.4	15	33.7	233.9	37.1
2	Ni_C	A.F.	33.7	40.4	9	93.9	50.0	51.9
3	?	A.F.	305.6	45.1	4*	69.3		
3	?	A.F.	101.8	2.4	3*	98.1		
4	Ni_C	A.F.	248.9	85.4	49	13.3	343.2	4.4
5	?	A.F.	260.9	−42.5	3*	48.2		
6	Ni_C	A.F.	21.4	56.3	35	13.2	21.2	53.4
4	?	A.F./T	356.7	53.4	−8/−2	8.5/19.9		
5	Ni_A	A.F./T	294.9	26.0	5	37.4	274.3	26.7
11	Ni_A	A.F.	295.2	52.3	19	10.8	295.7	24.4
11	Ha_{Al}†	A.F.			39	7	255.0	4.0
7								
11	Ha_B	A.F./T	129.0	17.1	18	13.7	247.0	37.0
8			147.2	29.3	−8/−2	14.1/29.3		
10	Ha_C	A.F./T	261.0	38.2	38	10.9	109.2	5.0
			227.5	65.3	−5/−1	22.0/41.0		
5	Ni_A	A.F./T			40	12.3	272.8	20.9
3	Ni_B	A.F./T			41	20.0	234.8	43.2
5	Ni_C	A.F./T			6	33.9	15.6	35.9
6	$Ni_A + Ha_{Al}$	A.F./T			28	13.0	269.1	17.5
4	$Ni_B + Ha_B$	A.F./T			51	13.0	232.8	45.0
6	$Ni_C + Ha_C$	A.F./T			4*	40.0	4.7	28.1

$^{40}Ar/^{39}Ar$ RESULTS

Table 2 and Figs. 8–12 summarize the analytical results for the hornblende, biotite, potassium feldspar and plagioclase from Nimba County, and the plagioclase from the Harper amphibolite. The analytical results from hornblende and biotite in the Harper amphibolite were published by Onstott *et al.* (1984).

Northeastern Liberia

The hornblende from the site 227 amphibolite yields an integrated date of 2174 ± 4 Ma, which agrees within error with the 2220 ± 60 Ma hornblende date reported by Hedge *et al.* (1975) for the metanorite at site 235. The step-heating spectra reveals, however, a profile consistent with that expected for contamination by excess ^{40}Ar (Harrison and McDougall 1981), with initially high ages for the low temperature fractions, dropping to a 2040 ± 4 Ma plateau over 60% of the $^{39}Ar_K$ release at higher temperature steps (Fig. 8).

The biotite from the site 227 biotite–quartz–feldspar schist yields a 1468 ± 2 Ma integrated date. The shape of the step-heating profile is reminiscent of that expected for ^{40}Ar loss during an episodic thermal event (Turner 1969), climbing from ~900 Ma date at low temperatures to a plateau date of 1508 ± 2 Ma (Fig. 9). Utilizing the least squares fitting procedure of Hall and York (1982) to determine the three parameters of such a model, we obtained a primary cooling age of 1510 ± 2 Ma, in close agreement with the plateau date, a 569 ± 11 Ma estimate for the timing of the thermal episode, and an estimated 7% loss of ^{40}Ar during that thermal event.

Potassium feldspar from the same biotite–quartz–feldspar schist yields an integrated date of 1348 ± 2 Ma. Like the biotite from the same sample, the $^{40}Ar/^{39}Ar$ release spectrum exhibits a pattern characteristic of ^{40}Ar loss, climbing from ~650 Ma dates at low temperatures to ~1500 Ma dates at the highest temperatures (Fig. 10). A least squares fit of 80% of the spectra yields a primary cooling age of 1512 ± 2 Ma, a 667 ± 12 Ma estimate for the timing of the thermal episode, which caused a 24% loss of ^{40}Ar.

An anomalously old integrated date of 3625 ± 6 Ma was obtained for the plagioclase from the site 227 amphibolite. Its saddle-shaped spectrum (Fig. 11), however, is characteristic of excess argon contamination (Lanphere and Dalrymple 1976, Harrison and McDougall 1981). The minimum ages of the saddle occur between 22 and 73% of the ^{39}Ar release and yield a minimum plateau date of 1446 ± 12 Ma. This minimum portion of the spectrum suggests that the plagioclase was completely reset by a thermal episode at this time, although diffusive gain of argon may have occurred later.

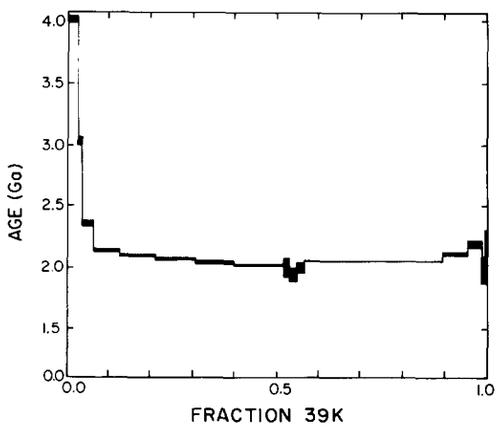


Fig. 8. $^{40}Ar/^{39}Ar$ spectra for the site 227 amphibolite hornblende. Error bars for each temperature step represents ±1 S.D. not including the error in the *J*-value.

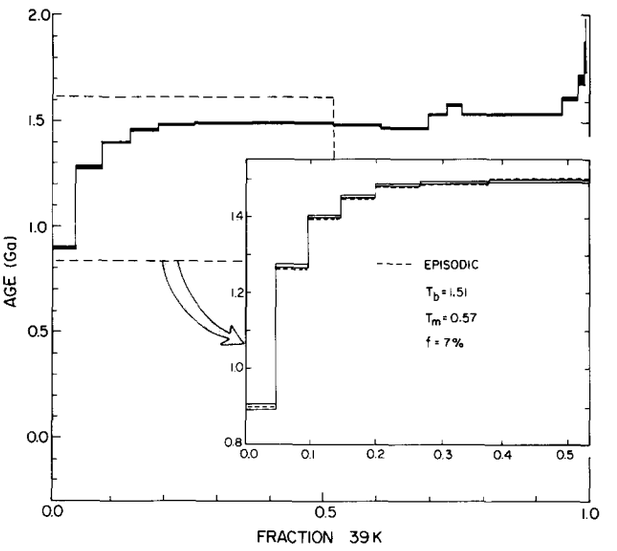


Fig. 9. $^{40}Ar/^{39}Ar$ spectra for the site 227 biotite. Inset compares the age spectra with a theoretical diffusive loss model (dashed line) for the parameters shown.

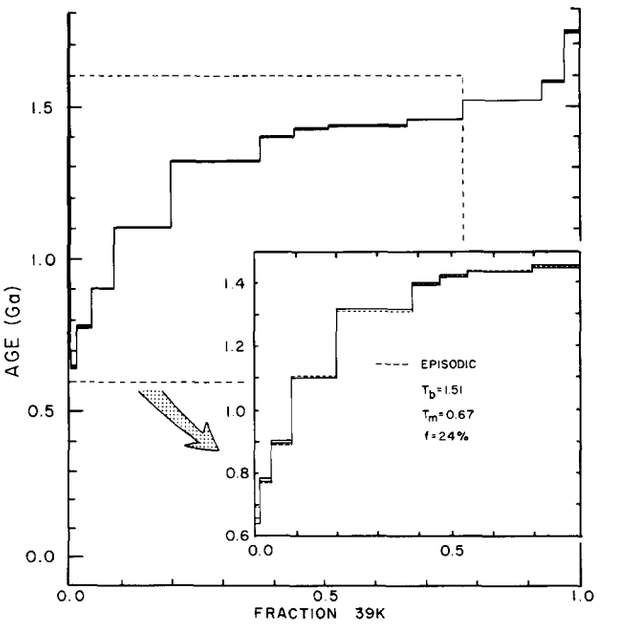


Fig. 10. $^{40}Ar/^{39}Ar$ spectra for the site 227 potassium feldspar. Inset compares the age spectra with a theoretical diffusive loss model (dashed line) for the parameters shown.

Table 2. $^{40}Ar/^{39}Ar$ analytical data

T(°C)	f_1	f_2	Atmos (%)	^{40}Ar ($\times 10^{-6}$ cm^3 g^{-1} STP)	f(39)	$^{37}Ar_{Ca}/^{39}Ar_K$	Date (Ma)
227 hornblende (70/120)*							
938	−0.021	0.086	5.2	0.29	0.025	32.4	4032 ± 26
950	−0.015	0.027	16.9	0.28	0.013	23.1	3030 ± 38
962	−0.017	0.069	14.7	0.32	0.026	26.0	2362 ± 19
975	−0.004	0.041	7.9	0.30	0.061	5.49	2137 ± 12
987	−0.008	0.152	4.9	0.25	0.087	12.8	2097 ± 10
999	−0.009	0.159	4.8	0.27	0.096	13.1	2070 ± 7
1011	−0.008	0.124	5.8	0.31	0.091	11.8	2040 ± 9
1023	−0.006	0.110	5.0	0.33	0.119	8.45	2024 ± 7
1036	−0.036	0.130	19.1	0.16	0.013	55.3	1999 ± 73
1049	−0.013	0.055	18.8	0.19	0.016	19.5	1946 ± 49
1062	−0.018	0.086	16.3	0.19	0.019	27.2	1999 ± 45
1075	−0.007	0.113	5.9	1.13	0.329	10.9	2056 ± 3
1100	−0.014	0.077	13.7	0.55	0.060	21.0	2120 ± 11
1125	−0.008	0.022	22.1	0.60	0.035	12.7	2198 ± 28
1150	−0.028	0.006	51.1	0.36	0.007	42.9	1974 ± 105
1200	−0.027	0.139	41.3	0.24	0.006	41.2	2176 ± 150
Mass = 0.0455						Integrated date =	2174 ± 4
						Plateau date (1011–1075°C) =	2040 ± 4
227 biotite (35/70)							
510	0.0011	−0.0022	57	17.4	0.046	1.70	898 ± 7
550	−0.0012	0.0057	30	9.52	0.050	1.81	1272 ± 6
575	0.0002	−0.0016	21	7.16	0.051	0.36	1400 ± 4
600	0.0005	−0.0056	15	5.33	0.054	0.08	1451 ± 6
625	−0.0010	0.0169	10	4.27	0.068	0.02	1485 ± 2
650	0.0007	−0.0171	8	5.08	0.105	0.01	1493 ± 2
675	0.0003	−0.0081	6	6.08	0.156	0.004	1496 ± 2
700	−0.0004	0.0110	7	3.47	0.088	0.006	1481 ± 3
733	0.0003	−0.0065	8	4.54	0.089	0.004	1462 ± 3
766	−0.0010	0.0102	15	3.53	0.035	0.002	1532 ± 4
855	−0.0005	0.0031	19	4.04	0.028	0.007	1572 ± 11
1028	0.0004	−0.0084	7	8.92	0.188	0.006	1534 ± 1
1050	−0.0001	0.0010	15	3.27	0.028	0.018	1609 ± 8
1100	−0.0017	0.0048	29	3.24	0.011	0.025	1702 ± 23
1150	−0.0026	−0.0016	61	3.93	0.003	0.400	1802 ± 72
1190	−0.0873	−0.0751	89	5.77	0.001	1.340	2242 ± 236
Mass = 0.0174						Integrated date =	1468 ± 2
						Plateau date (625–1050°C) =	1508 ± 2
227 microcline (70/120)							
500	−0.0001	0.0011	7	0.49	0.003	0.174	2472 ± 10
550	0.0001	0.0025	20	0.48	0.005	0.181	685 ± 7
600	0.0000	0.0017	10	0.49	0.013	0.047	647 ± 6
650	−0.0000	0.0003	3	0.43	0.027	0.003	779 ± 2
700	−0.0000	0.0003	2	0.54	0.048	0.003	900 ± 2
750	0.0000	0.0003	17	0.46	0.004	0.033	1015 ± 14
800	0.0000	0.0024	1	0.51	0.109	0.008	1101 ± 6
850	0.0000	0.0006	11	0.37	0.004	0.044	1244 ± 5
900	−0.0000	0.0003	0	0.68	0.178	0.001	1316 ± 1
940	0.0000	0.0002	0	0.34	0.068	0.001	1395 ± 1
980	−0.0000	0.0022	1	0.48	0.068	0.012	1422 ± 1
1020	−0.0000	0.0005	1	0.78	0.156	0.002	1433 ± 1
1060	0.0000	0.0007	1	0.66	0.108	0.003	1451 ± 1
1100	−0.0000	0.0001	11	0.54	0.005	0.009	1448 ± 12
1140	0.0000	0.0006	1	1.90	0.157	0.005	1513 ± 1
1180	0.0000	0.0002	2	1.22	0.047	0.004	1573 ± 1
Mass = 0.0295						Integrated date =	1348 ± 2
227 plagioclase (70/120)							
500	−0.0112	0.0048	3	0.13	0.021	17.3	7481 ± 38
550	0.0043	0.0037	7	0.11	0.023	6.6	5288 ± 44
600	−0.0034	0.0078	7	0.07	0.028	5.3	4292 ± 36
650	−0.0003	0.0008	10	0.10	0.044	0.4	3386 ± 26
700	−0.0027	0.0074	19	0.11	0.047	4.1	2388 ± 46
750	−0.0038	0.0144	22	0.12	0.063	5.8	1860 ± 24
800	−0.0015	0.0040	34	0.14	0.059	2.2	1486 ± 41
850	−0.0009	0.0048	24	0.17	0.117	1.4	1468 ± 24
900	−0.0026	0.0181	21	0.14	0.125	4.0	1439 ± 25
940	−0.0022	0.0134	24	0.12	0.092	3.3	1338 ± 28
980	−0.0025	0.0149	27	0.14	0.108	3.8	1501 ± 22
1020	−0.0007	0.0024	23	0.15	0.078	1.0	1894 ± 42
1060	−0.0043	0.0113	18	0.15	0.061	6.6	1569 ± 21
1100	0.0057	0.0063	17	0.19	0.045	8.7	3485 ± 22
1140	−0.0065	0.0011	15	0.25	0.033	9.9	4688 ± 33
1180	0.0187	−0.0034	9	0.12	0.013	0.3	5910 ± 62
1600	0.0013	0.0058	1	0.06	0.041	2.0	7091 ± 27
Mass = 0.1162						Integrated date =	3625 ± 6
						Plateau date (800–980°C) =	1446 ± 12

Continued overleaf.

Table 2. Continued

T(°C)	f_1	f_2	Atmos (%)	^{40}Ar ($\times 10^{-6}$cm^{3}g^{-1}STP)	f(39)	$^{37}Ar_{Ca}/^{39}Ar_K$	Date (Ma)
239 plagioclase (70/120)							
550	−0.004	0.002	5	5.24	0.018	5.60	6461 ± 40
600	−0.006	0.010	6	1.66	0.014	9.00	4905 ± 20
650	−0.006	0.022	16	0.44	0.018	9.46	2462 ± 38
700	−0.009	0.042	21	0.36	0.027	13.64	1712 ± 35
750	−0.012	0.073	19	0.46	0.038	18.19	1607 ± 34
800	−0.014	0.170	10	0.84	0.070	20.87	1601 ± 14
835	−0.014	0.202	9	0.86	0.075	21.36	1559 ± 11
866	−0.014	0.146	12	0.85	0.077	21.04	1513 ± 25
900	−0.013	0.179	10	0.72	0.069	20.38	1465 ± 20
935	−0.012	0.116	15	0.67	0.065	19.08	1447 ± 17
966	−0.011	0.069	21	0.55	0.052	17.34	1462 ± 19
1000	−0.010	0.045	25	0.53	0.047	15.73	1541 ± 15
1050	−0.010	0.027	30	0.59	0.045	15.62	1715 ± 19
1100	−0.010	0.020	26	0.92	0.045	15.67	2220 ± 21
1152	−0.011	0.012	15	2.68	0.047	16.49	3715 ± 9
1200	−0.012	0.010	10	6.83	0.065	19.05	4709 ± 9
1300	−0.014	0.143	2	14.50	0.193	21.32	4151 ± 2
1450	−0.014	0.055	4	3.47	0.032	20.77	4755 ± 11
1600	−0.013	−0.007	33	0.27	0.003	20.60	4508 ± 144
Mass = 0.1026						Integrated date =	3254 ± 3
						Plateau date (805–1000°C) =	1515 ± 7

Standard J-value = 0.02202 ± 0.00031†

* Sieve mesh sizes of separate.

† Weighted mean of three fusions of irradiation standard Zartman hornblende, 3 gr, which as a K–Ar date of $f_1 = 1/[1 - 37/39)_{Ca}/(37/39)_M]$, and $f_2 = f_1[1 - 36/39)_{Ca}/(36/39)_M]$, where $()_{Ca}$ = isotope ratio of argon extracted from irradiated calcium salts and $()_M$ = isotope ratio of argon extracted from irradiated unknown. ^{40}Ar = radiogenic ^{40}Ar f(39) = cumulative fraction of ^{39}Ar released in each step.

Date (Ma) = the date calculated using the following decay constants: $\lambda_\epsilon = 0.581 \times 10^{-10}\,y^{-1}$; $\lambda_\beta = 4.961 \times 10^{-10}\,y^{-1}$; $^{40}K/K = 0.01167$ atom % (Steiger and Jager 1977). The quoted error is one standard deviation and does not include the error in the J-value or the standard error.

Integrated date = the date and error calculated from the sum total gas from all steps; the error includes the error in J-value.

Plateau date = the date and error calculated from the sum total gas from those steps, the dates of which fall within 2 S.D. of each other; the error includes the error in J-value.

Mass = grams.

Southern Liberia

The plagioclase separate from the Harper amphibolite also yields an anomalously old integrated date of 3254 ± 3 Ma with a saddle shaped spectra (Fig. 12). The minimum portion of the saddle, which lies between 11 and 55% ^{39}Ar released, yields an integrated date of 1515 ± 7 Ma.

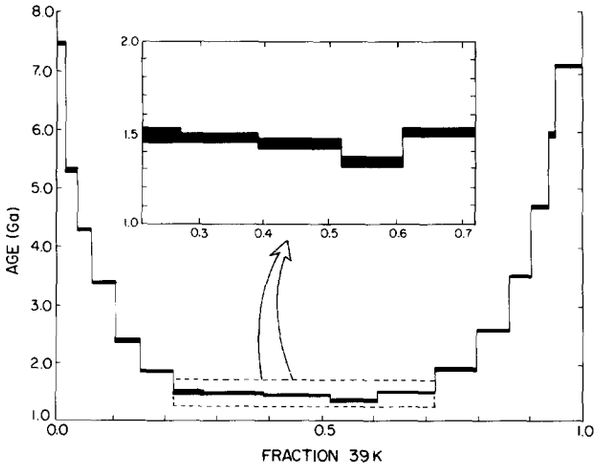


Fig. 11. Saddle-shaped ^{40}Ar/^{39}Ar spectra for the site 227 plagioclase. Inset enlarges the minimum portion of the saddle.

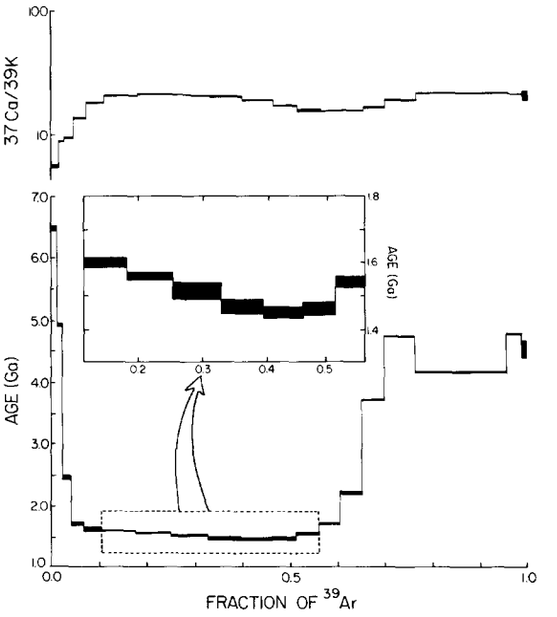


Fig. 12. Saddle-shaped ^{40}Ar/^{39}Ar spectra (bottom) and ^{37}Ar/^{39}Ar spectra (top) for the site 239 plagioclase. Inset enlarges the minimum portion of the saddle.

DISCUSSION

Thermal history

Hurley *et al.* (1971) reported ~2.7 Ga Rb–Sr ages and Hedge *et al.* (1975) reported ~2.7 Ga K–Ar hornblende ages from rock units in northern Liberia, where apparently the disturbing effects of Eburnean metamorphism are less severe. Since temperatures of the order of 450–550°C (Harrison 1981) are required to reset the argon isotopic clock in hornblende and since no secondary greenschist facies overprinting of the high grade mineral assemblages was observed in thin section, we propose that the recorded metamorphic facies reflect Eburnean metamorphism and that the $^{40}Ar/^{39}Ar$ hornblende date as such records post-Eburnean uplift and cooling.

The consistency of the 1.5 Ga $^{40}Ar/^{39}Ar$ dates from biotite and feldspars from both northeastern and southern Liberia indicates that a widespread, regional thermal event took place at that time, the origin of which is unknown. The other ~1.5 Ga K–Ar mineral dates reported for western Liberia (Stobernack 1970) and for Ivory Coast (Maluski and Schaeffer 1981) may be related to this event. In order to cause complete argon loss from the Nimba biotite and K-feldspar, temperatures must have reached at least 300°C.

The $^{40}Ar/^{39}Ar$ step-heating profiles for the biotite and microcline both indicate the occurrence of a third thermal episode between ~570 and ~670 Ma, which is most plausibly related to the Pan-African orogeny. Because the biotite and microcline lost only a fraction of their argon, the maximum temperature attained during the event was probably no more than 250°C. The apparent absence of this 570–670 Ma dates from the saddle-shaped, excess argon, $^{40}Ar/^{39}Ar$ spectra for both plagioclase spectra suggests that either (1) the blocking temperature of the plagioclase is greater than that of biotite and microcline or (2) during this thermal episode the partial fluid pressure of ^{40}Ar was sufficiently high to cause diffusive gain of ^{40}Ar in the K-poor plagioclase, but not high enough to prevent diffusive loss of ^{40}Ar from the K-rich biotite and microcline.

Magnetization history

The predominant component, Ni_A, is carried by fine-grained magnetite inclusions in orthopyroxenes that appear to have been formed prior to or above the hornblende blocking temperature (Fleet *et al.*, 1980), the 2.04 Ga age is a minimum estimate for the age of Ni_A. This hypothesis is consistent with the apparent similarity of the Ni_A directions with that of Ha_{A1} from the Harper amphibolite (Table 1), which yielded a 1.96 Ga $^{40}Ar/^{39}Ar$ hornblende age. We also note that the close similarity in the paleomagnetic poles for the two regions, Nimba and Harper, which are separated by ~300 km, indicate that the effect of any post-magnetization structural reorientations are relatively minor.

Because the $^{40}Ar/^{39}Ar$ data indicate that temperatures of ~300°C were attained at ~1.5 Ga, it is unlikely that component NI_B which in part is carried by pyrrhotite with ~320°C Nèel temperature (Craig and Scott 1974) could possibly predate this thermal episode. The Harper amphibolite also carries a less stable component, Ha_B, which is similar in direction to Ni_B, and its plagioclase also yields a ~1.5 Ga $^{40}Ar/^{39}Ar$ date. Based upon this apparent coincidence in the $^{40}Ar/^{39}Ar$ results on feldspars from both regions and the similarity in their paleomagnetic poles, we propose that remanent components Ni_B and Ha_B are both secondary magnetizations, possibly pTRMs acquired during the ~1.5 Ga thermal episode.

Alternatively, one could interpret the Ni_B and Ha_B components as representing poorly resolved remanent overprints induced by a geomagnetic field opposite in polarity to that recorded by components Ni_A and Ha_A, presumably during slow (~1°C/Ma), post-Eburnean cooling of the Nimba and Harper rock units. Despite their occurrence as distinct linear segments on orthogonal plots and the statistically significant difference in their average pole positions (see Table 1), one cannot convincingly preclude this interpretation in favor of that outlined above solely on the basis of the paleomagnetic data. Component Ni_B, however, is carried in some samples by pyrrhotite with a ~320°C Nèel temperature, whereas, component Ni_A is carried predominantly by single domain magnetite with a ~580°C Curie temperature. Furthermore, $^{40}Ar/^{39}Ar$ analyses of hornblende with a ~500°C blocking temperature yielded a ~2.0 Ga date, whereas biotite with a ~300°C blocking temperature yielded a ~1.5 Ga date. Accepting these additional petrographic and isotopic constraints at face value, we prefer to believe that components Ni_B and Ni_A are separated in age by approximately 500 Ma, and do not simply represent a mixture of incompletely resolved magnetizations, which recorded reversals of the geomagnetic field during a single cooling episode.

The paleomagnetic poles corresponding to the 'C' components from the Nimba rocks and Ha_C from the Harper amphibolite fall within the general region of late Precambrian to Ordovician poles for Gondwanaland (Klootwijk 1980). This suggests that the 'C' component was acquired during the ~570–670 Ma low-temperature thermal episode detected in the $^{40}Ar/^{39}Ar$ analyses of biotite and potassium feldspar from the Nimba schist. The scatter in 'C' component poles may reflect a prolonged period of acquisition by a combination of thermal and chemical processes or by minor structural reorientation following magnetization during the Pan-African orogeny. Despite the scatter, the average pole does coincide closely with a reliably dated Pan-African age pole (see Table 3) from the Adima diorite (Morel 1981).

Although these correlations of the secondary magnetization with low temperature thermal events are highly speculative and require further combined $^{40}Ar/^{39}Ar$ and paleomagnetic studies of other rock units in Liberia, the position of the ~2.0 Ga paleomagnetic pole for Liberia, and hence the West African Shield, has been determined

with sufficient reliability to compare it with similar age poles from other Precambrian shields.

Tectonic implications

Table 3 lists previously published paleomagnetic poles for the West African Shield and the radiometric dates reported for these or related rock units. These poles are plotted along with the results of this study in Fig. 13. Our results fall within a cluster (dashed oval in Fig. 13) of other poles from the Aftout diotite of Algeria (EP_C and EP_E in Fig. 13; McElhinny and Cowley 1980) and a diabase dike in Ivory Coast (ICD in Fig. 13; Piper and Lomax 1973) all reflecting post-Eburnean magnetizations from 2.05 to 1.7 Ga. As pointed out by Onstott *et al.* (1984) this cluster is not consistent with the 2.05–1.8 Ga paleomagnetic data from the Kalahari Shield (hachured arrow in Fig. 13), indicating that the present-day configuration of the Africa cratons did not exist 2 Ga ago.

The Aftout gabbro (AF) and Eglab quartz diorite (YED) poles, however, do appear to be consistent with the paleomagnetic data from the Kalahari Shield (Morgan 1985), but their ages of magnetization have yet to be reliably determined. The paleomagnetic poles from the greenstone rocks and dikes of Ghana were originally interpreted by Piper and Lomax (1973) as being consistent with the paleomagnetic data from the Kalahari Shield assuming that their magnetizations were acquired prior to the Eburnean orogeny (i.e. ~2.1–2.2 Ga). The radiometric data (see Table 1), their relative stratigraphic ages, their lack of stability to A.F. demagnetization, and their questionable fold test (see Table 3, footnote), however, do not support a pre-2.0 Ga age.

We also note that the paleomagnetic data from the Kalahari Shield indicate very little apparent polar wander between 2.05 and 1.8 Ga. If the published age interpretation for the West African Shield paleomagnetic poles, which are widely scattered, are accepted at

Table 3. Precambrian paleomagnetic data for the West African Shield

Formation	Abbv.*	No. of sites (samples)	Lat. (°N)	Long. (°E)	K (κ)	A_{95} (α_{95})	Possible magnetization ages†	Refs.
Liberia								
Nimba + Harper	Ni_A	6 (38)	18	269	28	13	1894 ± 2 ⩽ t ⩽ 1964 ± 4, 2050 ± 6 AA hb	1,2
Nimba + Harper	Ni_B	2 (16)	45	243	12	79	⩽1510 ± 10 AA b, k	2
							⩾1450 ± 10 AA pl	1
							⩾1450 ± 10 AA pl	2
Algeria								
Ecglab Plutons C	EP_C	4 (24)	5	278	77	11		
Ecglab Plutons E	EP_E	6 (40)	6	270	201	5	1998 ± 80 K b ⩽ t ⩽ 2020 ± 54 R	3,4
Aftout Gabbro	AF	5 (26)	29	55	146	6		
Yetti-Ecglab Q. Dio.	YED	3 (18)	7	9	32	22	⩽1935 ± 50 R	3,4
Yetti Dikes	YD	7 (34)	41	31	4	33	⩽1963 ± 38 R	3,5
Guelb El Hadid dikes	GEH	9 (53)	35	285	10	18	1339 – 1754 K	3,6
Ghana								
Obuasi greenstone	OBG‡	5 (74)	50	282	(21)	(19)	1800 – 2150 K m ⩽ t	7,8
							t ⩽ 1930 K hb, 2060 R	8,9
Obuasi dike	OBD§	1 (14)	56	249	(21)	(11)		7
Tarkwa dikes	TD‖	5 (29)	53	197	(14)	(21)	*in situ*	7
			53	216	(32)	(14)	*struc. corr.*	7
Ivory Coast								
Dike	ICD	2 (10)	11	282	(3)	(39)	1739 ± 130 K	7,10
Kanangono kimberlite	KK	1 (7)	63	67	(28)	(10)	1479 ± 54 R b–Wr iso	11,12
Mauritanie								
Groupe de Char	I2I	5 (23)	54	32	(160)	(6)	998 ± 34 R cl	13,14
	I2S	5 (25)	67	258	(36)	(11)	560 – 675 K cl	13,14
Mali								
Adma diorite	AD	13 (62)	33	345	(22)	(9)	558 – 645 R b ⩽ t ⩽ 616 ± 13 U	15,16,17

* Abbreviations used in Fig. 13.

† All radiometric ages have been corrected to the decay constants of Steiger and Jager (1977). The uncertainties reflect ±1 S.D. and do not necessarily represent the uncertainty in the age of the magnetization. R = Rb/Sr, K = K/Ar, U = U/Pb zircon, AA = $^{40}Ar/^{39}Ar$, b = biotite, mu = muscovite, hb = hornblende, cl = clay, and unless otherwise indicated the ages were derived from total rock analyses.

‡ The Obuasi greenstone pole is based on NRM directions as the samples were unstable to A.F. demagnetization, and as the pole lies near poles derived from Jurassic swarms, it may represent an overprint.

§ The Obuasi dike is undated and as the pole lies very close to those derived from Jurassic dike swarms, it may be a Jurassic dike.

‖ The structural correction was applied by rotating one dike to vertical, and not by correcting bedding to horizontal. Consequently, it is not a true fold test.

References: 1. Onstott *et al.* 1984; 2. this study; 3. McElhinny and Cowley 1980, 4. Lasserre *et al.* 1970, 5. Sabate 1978, 6. Sabate and Lomax 1975, 7. Piper and Lomax 1973, 8. Tugarinov 1967, 9. Kolbe and Pinson 1967, 10. Knopf 1967, 11. Strangway *pers. comm.* 1982, 12. Bardet and Vachette 1966, 13. Morris and Carmichael 1977, 14. Clauer 1981, 15. Morel 1981, 16. Ducrot *et al.* 1979, 17. Davison 1981.

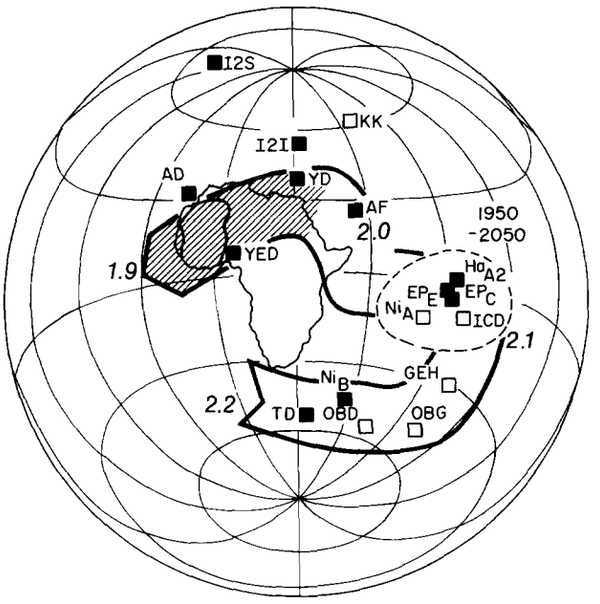


Fig. 13. Equal area stereo projection showing the Precambrian paleomagnetic poles for the West African Shield. The abbreviations correspond to those given in Table 3. Solid squares are poles and open squares are antipoles. Dashed oval encloses the reliably dated ~2.0 Ga poles of this study and those listed in Table 3. The APW path of McElhinny and McWilliams (1977) is outlined (thick lined arrow). The only portion of that path on which the paleomagnetic poles from the Kaapvaal Craton lie is hachured.

face value, then rapid apparent polar wander is required for the same period of time.

CONCLUSIONS

Combined paleomagnetic and $^{40}Ar/^{39}Ar$ analyses of metamorphic rocks from northeastern Liberia detected the presence of three components of magnetization, Ni_A, Ni_B and Ni_C, which were acquired as TRMs, pTRMs and CRMs, during three thermal episodes at 2.0, 1.5 and 0.6 Ga, respectively. The paleomagnetic pole for the ~2.0 Ga old Ni_A component coincides with the ~1.95 Ga old Ha_{A1} paleomagnetic pole from southern Liberia, and agrees with other poles of similar age from the West African Shield, but not with coeval poles from the Kaapvaal Craton. This data suggests some form of displacement between the West African and Kalahari shields has taken place since 2.0 Ga.

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